

Lab 06 - Logic and Proof

Objective

Solving exercises from the textbook in chapter 1.6-1.8

Current Lab Learning Outcomes (LLO)

By completion of the lab, the students should be able to:

1. Students will understand Methods of proof.
2. They will be able to solve shorter/easier or longer / harder problems given in the textbook.

Lab Requirements

Students allowed using their lecture notes in the lab in order to solve the exercises.

Lab Assessment

- 1- Divide students to groups and let them to solve the given example.
- 2- Discuss the answers with the groups and write on board the optimal solution.

Lab Description

In this lab, the following exercises are going to be solved and explained to them:

- 1 Use a proof by contradiction to prove that the sum of an irrational number and a rational number is irrational.
- 2 Show that these statements about the integer x are equivalent:
(i) $3x + 2$ is even, (ii) $x + 5$ is odd, (iii) x^2 is even.
- 3 Show that if n is an integer and $n^3 + 5$ is odd, then n is even using
 - a) a proof by contraposition.
 - b) a proof by contradiction.
- 4 Prove the following statement by contradiction if n^2 is odd, then n is odd.
- 5 Show that at least ten of any 64 days chosen must fall on the same day of the week.
- 6 Prove that If you are studying CPCS222 then you are human (domain is all people), what kind of proof did you use.
- 7 Show that if $x^2 \geq 0$ then $x^2 + 2x + 1 \geq 0$ (Domain is of all real numbers). what kind of proof did you use.
- 8 Show that if $x^2 + 1 < 0$ then $2x^2 + 5x + 1 = 0$ (Domain is of all real numbers). what kind of proof did you use.
- 9 Prove that if x and y are real numbers, then $\max(x, y) + \min(x, y) = x + y$. [Hint: Use a proof by cases, with the two cases corresponding to $x \geq y$ and $x < y$, respectively.]
- 10 Prove that there is no positive integer n such that $n^2 + n^3 = 100$.

- 11 Show that the implication If $x > 0$, then $x^2 + 5 > 0$. Where x is any real number.
- 12 Prove that there are no solutions in integers x and y to the equation $2x^2 + 5y^2 = 14$.
- 13 prove that the following implication is true, where x is a real number and n is a constant.

$$\text{If } (1+x)^n > 1 + \frac{nx}{1!} + \frac{n(n-1)x^2}{2!}, \quad \text{then } x^2 + 1 > 0.$$

what ever the value of x then $x^2 + 1 > 0$ is always true . So without considering p, q is always true.